Asteroseismology and the Radial Adiabatic Pulsation Equations

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Objectives

- Motivate the derivation of radial adiabatic pulsation
- Set up the equations from fluid mechanics applied to stellar p, ρ, and Φ gradients
- Perturb these equations to the first order
- Explain the radial adiabatic solution process
- Solve for a toy sun with uniform density

What is Asteroseismology?

Asteroseismology is the **study of oscillations in stars**. Its model of these oscillations helps us **make predictions about internal stellar structure from observed oscillations**, and vice versa.

The radial adiabatic model is a useful, simple model of stellar oscillations. It asks us to consider: **what happens if we radially perturb a star in equilibrium while conserving entropy?**

We use more complex stellar models that allow for non-radial, non-adiabatic pulsation to model variable stars like Cepheids and δ Scutis. These are solved using numerical methods for the equations of motion, and provide **independent constraints on stellar properties**.

Stellar Equations at Equilibrium

The adiabatic radial model comes from the **mass**, **momentum**, and **energy conservation** equations, subject to **gravity** and **adiabaticity**.

$$\text{Mass: } \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0 \tag{1}$$

$$\text{Momentum: } \rho_0 \frac{\partial v}{\partial t} = -\frac{\partial p_0}{\partial r} - \rho_0 \frac{\partial \Phi}{\partial r} = \underset{\text{eq}}{0} \tag{2}$$

$$\text{Energy: } \frac{\mathrm{d}p}{\mathrm{d}t} = \frac{\Gamma_1 p}{\rho}\frac{\mathrm{d}\rho}{\mathrm{d}t}, \quad \Gamma_1 = \left(\frac{\partial \ln p}{\partial \ln \rho}\right)_{\text{s}} \tag{3}$$

$$\text{Gravity: } f_{\text{g}} = -\rho_0 \frac{\partial \Phi}{\partial r}, \quad \nabla^2 \Phi = 4\pi G \rho_0 \tag{4}$$

Perturbation Analysis

We can derive solutions for the radial adiabatic pulsation by starting with a **radial Lagrangian perturbation** of δr. Moving a parcel of gas from r_0 to $r_0 + \delta r$, the Lagrangian δp of the parcel is:

$$\underbrace{\delta p(r)}_{\text{Lagrangian}} = p(r_0 + \delta \mathbf{r}) - p_0(\mathbf{r_0}) =$$
$$p(\mathbf{r_0}) + \delta \mathbf{r} \cdot \nabla p_0 - p_0(\mathbf{r_0}) = \underbrace{p'(\mathbf{r_0})}_{\text{Eulerian}} + \delta \mathbf{r} \frac{\mathrm{d}p_0}{\mathrm{d}r} \tag{5}$$

This is the **Eulerian** perturbation p' plus the radial perturbation of the pressure gradient $\frac{\mathrm{d}p_0}{\mathrm{d}r}$.

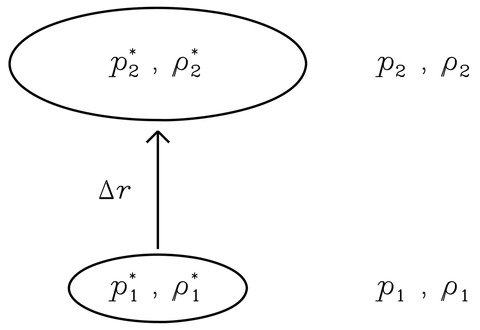


Figure 1: Consider a parcel going from $\mathbf{r_1}$ to $\mathbf{r_2}$, where $*$ denotes the new state. $p_2^* - p_1$ is δp; $p_2^* - p_2$ is p'. Figure from [1].

Boundary Conditions

Eq. 8 relates eigenfrequency ω^2 and eigenfunction $\zeta(r,t)$ to the equilibrium profiles r_0, p_0, ρ_0, and Γ_0. For the adiabatic radial case, **this is tractable** with boundary conditions: a perturbation at $r = 0$ is linear with radius, and $r = R$ is a free surface.

$$\lim_{r \to 0} \zeta(r) = \text{const} \tag{6}$$

$$\delta p(R) = p'(R) + \xi_r(R) \left.\frac{\mathrm{d}p_0}{\mathrm{d}r}\right|_R = 0. \tag{7}$$

Solving for the Sun

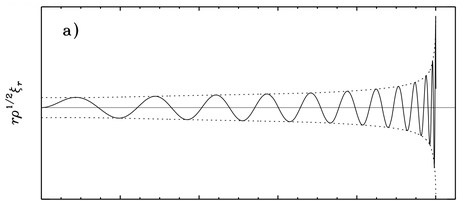


Figure 2: A high-order ($n = 23$) radial Solar eigenfunction from $r = [0, R_\odot]$ at $\nu = 3310$ μHz (5 mins). Figure from [1].

Using equilibrium profiles and Sturm-Liouville theory, we can solve for the eigenfrequencies in Eq. 8. The wave above is a high-frequency mode; the y axis $y = r\rho^{1/2}\xi_r$ is the square root of the integrand of the mode inertia ε, where average energy $\bar{E} = 1/4\omega^2\varepsilon$.

Adiabatic Radial Pulsation Equation

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(r_0^4 \Gamma_{1,0} p_0 \frac{\mathrm{d}\zeta}{\mathrm{d}r}\right) + r_0^3 \zeta \frac{\mathrm{d}}{\mathrm{d}r}\left[(3\Gamma_{1,0} - 4)p_0\right] + \rho_0 \omega^2 r_0^4 \zeta = 0, \quad \zeta(r) = \frac{\xi_r(r)}{r}. \tag{8}$$

Perturbed Stellar Equations

$$\text{Mass: } \frac{\partial \rho'}{\partial t} + \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \rho_0 v\right) = 0. \tag{9}$$

$$\text{Momentum: } \rho_0 \frac{\partial^2 \xi_r}{\partial t^2} = -\frac{\partial p'}{\partial r} - \rho' g_0 - \rho_0 \frac{\partial \Phi'}{\partial r} \tag{10}$$

$$\text{Energy: } p' + \xi \frac{\mathrm{d}p}{\mathrm{d}r} = \frac{\Gamma_{1,0} p_0}{\rho_0}\left(\rho' + \xi \frac{\mathrm{d}\rho}{\mathrm{d}r}\right) \tag{11}$$

$$\text{Gravity: } \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial \Phi'}{\partial r}\right) = 4\pi G \rho' \tag{12}$$

Simplified Uniform ρ Model

To solve for ω^2 without solving eigenfunctions, we can assume $\frac{\mathrm{d}\zeta}{\mathrm{d}r} = 0$. Further assuming uniform ρ, this is a star in **simple harmonic motion**:

$$\frac{\mathrm{d}}{\mathrm{d}r}[(3\Gamma_1 - 4)p] = -\rho r \omega^2 \tag{13}$$

Using $\frac{\mathrm{d}p}{\mathrm{d}r} = -\rho g(r)$ and $g(r) = Gm(r)/r^2$:

$$\omega^2 = -\frac{\mathrm{d}}{\mathrm{d}r}(3\Gamma_1 - 4)\frac{p}{\rho r} = (3\Gamma_1 - 4)\frac{Gm(r)}{r^3} \tag{14}$$

For $\Gamma_1 = 5/3$ (ideal H gas), $\omega^2 = Gm(r)/r^3$, giving stable oscillation with $\boxed{\text{P} = 2\pi\sqrt{r^3/Gm(r)}}$.

Checking with a Toy Sun ⊙

With $M = M_\odot$, $R = R_\odot$, we get $P = 10\,010\,\text{s}$. This is exactly equivalent to P_{orb} at the star's surface, given by the equation $P_{\text{orb}} = 2\pi\sqrt{R^3/GM_\odot}$. The **dynamical timescales of oscillation and surface orbit are equal** for this model.

Assumptions and Simplifications

- No rotation, no coulomb forces or viscosity
- Spherical symmetry of geometry and $\rho, p,$ and Φ distributions
- Non-adiabatic (heat transfer) effects can be ignored due to long timescales if ρ isn't tiny
- $\tau_\epsilon \gg P_{\text{puls}} \implies$ nuclear reactions can be ignored.
- Magnetic fields can be ignored; no MHD effects
- Pulsation heat transfer per cycle is negligible
- The exterior of the star at $r = R_\star$ is a free surface
- The star is a continuum where each $f(r)$ is defined for every $r \in [0, R_\star]$

References

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